

Bayesian Inference for Partial Orders from Tied Rank Order Data

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Summer project report

Abstract

Partial orders provide a natural framework for modeling ranking data when the underlying relations need not form a total order, allowing incomparabilities between elements to be represented explicitly. We develop a Bayesian framework for inferring an underlying partial order from ranking data with ties. Under an ignorability assumption on the observation mechanism, we derive a simplified posterior and develop an MCMC algorithm for posterior inference. We establish posterior consistency and introduce an asymptotically consistent consensus estimator. The method is evaluated through simulation experiments and applied to rankings of cognitive performance across 24 primate taxa.

1 Introduction

Ranking data arise in many settings, including human preferences over comparable items [Gormley and Murphy, 2009], outcomes of competitions between animals [Foerster et al., 2016], and rankings of individuals according to social status [Nicholls and Muir Watt, 2011]. Many statistical models have been proposed for analysing such data, including the Mallows model [Mallows, 1957] and the Plackett–Luce model [Plackett, 1975, Luce, 1959]. These models typically seek to recover a central ranking, represented by a total order, from ranking data observed on all or a subset of the objects of interest.

A total-order representation assumes that every pair of distinct objects is comparable. This assumption may be inappropriate in applications where the underlying relations are inherently incomplete. In particular, in some social hierarchies there may be pairs of individuals for which no meaningful ordering relation exists. A natural alternative is therefore to represent the underlying preference structure as a partial order, in which some pairs of objects may be incomparable. Statistical methods for recovering partial orders from ranking data are considerably less developed than methods based on total orders. Nicholls et al. [2024] develop a Bayesian framework in which an underlying social hierarchy is represented by a partial order and inferred from observed rank-order data.

Rank-order data may also contain ties, which can arise naturally as a consequence of the observation mechanism. Several methods have been developed for inferring an underlying total order from tied ranking observations; see, for example, Davidson [1970]. More recently, the hierarchical models developed in Nicholls et al. [2025], Li et al. [2026b] provide a framework that can be adapted to address tied observations in partial-order inference. In this report, we build on the Bayesian framework of Nicholls et al. [2024] to infer an underlying partial order from rank-order data with ties. Specifically, we extend the observation model to accommodate tied rankings, thereby allowing ties to occur in the observed data while retaining a partial order as the underlying preference structure. We further impose some assumptions on the observation model to improve the computational scalability of inference and develop an algorithm for posterior inference under the resulting model. Finally, we apply the proposed methodology to the tied rank-order data on primate cognitive abilities used by Johnson et al. [2002].

Section 1.1 introduces the background material and notation used throughout the report and provides a formal statement of the problem. Section 2 reviews existing prior and observation models for partial orders. In Section 3.1, we introduce our observation model for bucket orders, followed in Section 3.2 by an ignorability assumption that simplifies the resulting model. Section 3.3 derives the posterior distribution used to infer the unknown partial order h_0 and introduces an MCMC algorithm targeting this posterior. In Section 3.4, we establish asymptotic properties of the posterior and introduce an asymptotically consistent estimator of h_0 . Section 4 presents the results of our simulation experiments and applies the proposed methodology to the primate intelligence data of Johnson et al. [2002]. Finally, Section 5 concludes with a discussion of the results. The R code and data used to perform the analyses, together with additional discussion, are available on GitHub.

1.1 Background and Notations

A partial order on M , our object of inference, is a directed graph $h = (M, \prec_h)$, where $\prec_h$ is a set of relations on M , with elements of the form $i \prec_h j$ for some pairs of $i, j \in M$. The relation also has to be (i) transitive, i.e. $\forall i, j, k \in M (i \prec_h j \wedge j \prec_h k) \implies i \prec_h k$, and (ii) irreflexive, i.e. there is no $i \in M$ such that $i \prec_h i$. These properties also imply asymmetry, i.e. there are no $i, j \in M$ such that both $i \prec_h j$ and $j \prec_h i$. A partial order is also sometimes called a poset. If $\prec_h$ is in addition *complete*, that is $\prec_h$ satisfies $\forall i, j \in M, i \neq j : i \prec_h j$ or $j \prec_h i$, we call h a total order. Since the set of total orders on M is also in one-to-one correspondence with $\mathcal{P}_M$, we abuse notation by letting $\mathcal{P}_M$ be the set of total orders on M when there's no ambiguity. A bucket order $b = (M, \prec_b)$ is a poset such that there exists an ordered partition $(B_i)_{i=1}^{n'}$ of M , where $n' \leq n$ and B_i s are all non-empty, such that whenever $1 \leq k_1 < k_2 \leq n'$ one has $(i \in B_{k_1} \text{ and } j \in B_{k_2}) \implies i \prec_b j$, and in the case where $k_1 = k_2$ one requires $i \not\prec_b j$ and $j \not\prec_b i$. One can think of bucket orders as ties, where we put the tied actors in the same bucket. See Figure 1 for an illustration.

**use another partial order, this is an VSP

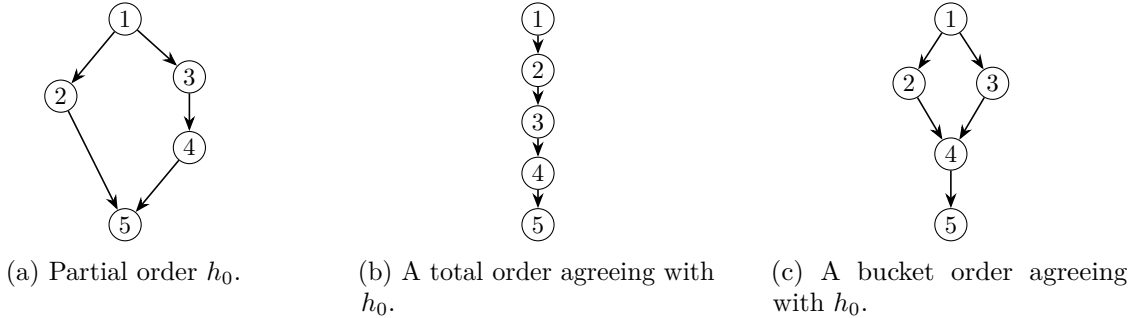


Figure 1: Illustration of partial, total, and bucket orders.

Given any 2 partial orders h_1 and h_2 on the same ground set M , we say h_1 disagrees with h_2 if there is some actors $i, j \in M$ s.t. $i \prec_{h_1} j$ (and) $j \prec_{h_2} i$, otherwise we say h_1 and h_2 agrees on one another. Given a partial order h on ground set M , a linear extension of h is defined to be any total order $l \in \mathcal{P}_M$ that agrees with h . We write $\mathcal{L}[h] \subset \mathcal{P}_M$ as the set of linear extension of h . Suppose we have N actors $(y_i, \dots, y_{N+i-1}) \subset M$, we will write $y_{i:(i+N-1)}$ as a shorthand of $(y_i, \dots, y_{N+i-1})$. The adjacency matrix of the poset h is the matrix $M^{(h)} \in \mathbb{R}^{n \times n}$ with definition $M_{i,j} = \mathbb{I}_{[i \succ_h j]}$. The Hamming distances between 2 posets h_1 and h_2 is defined by

$$\sum_{i,j=1}^n |M_{i,j}^{h_1} - M_{i,j}^{h_2}|.$$

The transitive closure of a DAG h is the directed graph $c(h)$ with the smallest transitive relation set $\prec_{c(h)}$ such that $\prec_h \subset \prec_{c(h)}$

We also call

$$\max(h) := \{i \in M : \nexists j \in M \text{ s.t. } i \prec_h j\}$$

the set of maximal element, or the max set of h . If the choice set is restricted from M to $S \subset M$, then preference are given by $h[S] = (S, \prec_h)$ with relations $i \prec_h j$ involving $i, j \in S$.

For the rest of the report, we will also denote $\mathcal{B}_M$ as the set of bucket orders on M , $\mathcal{P}_M$ as the set of total orders on M , and $\mathcal{H}_M$ as the set of partial orders on M . It should also be clear that we have $\forall n > 2, \mathcal{P}_M \subsetneq \mathcal{B}_M \subsetneq \mathcal{H}_M$.

The goal of this report is to infer an underlying partial order h_0 , based on N i.i.d. observations $\{(o^{(i)}, b^{(i)})\}_{i=1}^N$, where $o^{(i)} \subseteq M$ with $|o^{(i)}| \geq 2$ (2 since we need at least 2 elements in an observation to make meaningful comparisons) and $b^{(i)} \in \mathcal{B}_{o^{(i)}}$ for all $i = 1, 2, \dots, N$, which represents the bucket orders $b^{(i)}$ observed from a subset of elements $o^{(i)}$ on the ground set M .

2 Bayesian inference for Partial Orders with Ranking Data

2.1 Prior models

There has only been a few priors suggested for partial orders. One simple choice is the uniform prior, which has the distribution $\pi(\cdot) \sim \mathcal{U}[\mathcal{H}_M]$, meaning that every poset h of n actors is equally likely. However natural this prior might seem, it is in fact very informative over the depth, that is the quantity

$$d(h) = \max_{S \subseteq M} \{|S| : h[S] \in \mathcal{P}_S\}.$$

As an interesting result shown by Kleitman and Rothschild [1975], under $\pi(\cdot)$ one has

$$\mathbb{P}[d(h) = 3] \rightarrow 1 \quad \text{as } n \rightarrow \infty.$$

This is undesirable since the depth is often an object of interest that we would prefer the prior to be uninformative about, yet the uniform prior increasingly concentrates on partial orders of depth 3 as the number of actors grows.

Nicholls et al. [2025] suggested a prior using latent variables that can be tuned to be approximately uniform over depth. However, we will not consider both of them for simplicity in this report.

2.2 Noise-free observation model for total orders

Let $M = \{1, 2, \dots, n\}$ be the ground set of the underlying partial order $h \in \mathcal{H}_M$ that we are trying to learn about. Suppose for the purpose of demonstration that the data is $l = (l_1 \succ l_2 \succ \dots \succ l_n)$, which is a total order on M . This can be easily generalized to cases where the ground set of the total order is a proper subset of M . By noise-free, we mean that the model assigns probability 0 to any total order that is not a linear extension of h .

2.2.1 Uniform likelihood

One of the choices for a noise-free model is the uniform model:

$$p(l \mid h) = \frac{\mathbb{I}[l \in \mathcal{L}[h]]}{|\mathcal{L}[h]|}.$$

However, this is computationally intractable in general since computing $|\mathcal{L}[h]|$ is a $\#P$ -complete problem [Brightwell and Winkler, 1991]. In practice, exact counting is often only feasible when n is a few dozen, although the precise limit depends heavily on the structure of the partial order and the implementation [Talvitie and Koivisto, 2024]. We use `lecount()`, which implements a linear-extension counting algorithm proposed by Kangas et al. [2016]. One can avoid counting linear extensions using the single-variable exchange (see Appendix A.1). There are still some good reasons to use this model despite these computational difficulties. For instance, it can also be obtained as a limiting case of a partial-order Mallows model as the dispersion parameter tends to infinity (see Appendix A.1), providing an additional motivation for the uniform linear-extension likelihood.

2.2.2 The Frontier-Max (FM) likelihood

An alternative is to use a special case of the frontier-softmax likelihood model suggested by Li et al. [2026a], which is given by:

$$p(l \mid h) = \prod_{i=1}^n \frac{\mathbb{I}[l \in \mathcal{L}[h]]}{|\max(h[l_{i:n}])|}.$$

The model is motivated by an assessor realizing l from h by sequentially selecting one element at a time from the maximal elements of the poset. We assume the assessor start with an empty queue, and they choose the element in the queue one by one, by moving uniformly at random element from the maximal element of the poset induced by the actors that hasn't been added to the queue, until the element from the ground set is exhausted. An illustration is given in Figure 2.

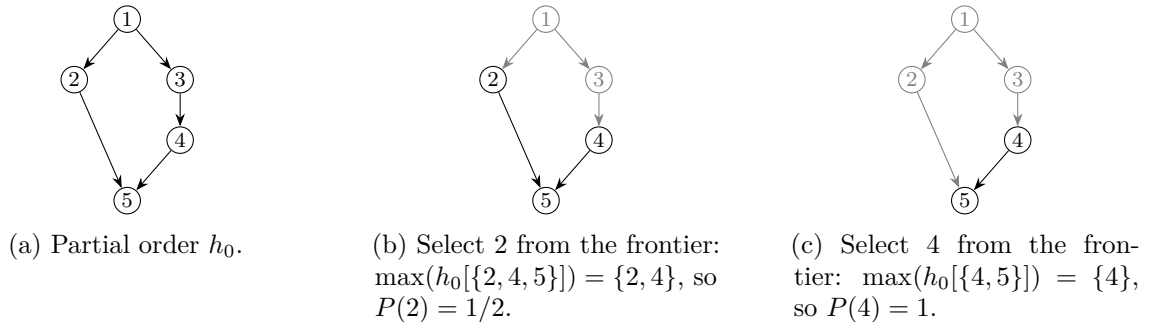


Figure 2: Illustration of the FM selection process for the linear extension $l = (1, 3, 2, 4, 5)$. After 1 and 3 have been selected, the frontier is $\{2, 4\}$, so 2 is selected with probability $1/2$. Removing 2 leaves the frontier $\{4\}$, so 4 is selected with probability 1. Grey nodes denote actors that have already been selected.

The model is tractable and simple. Furthermore, we have also shown that its MLE is the same as the one given by the uniform likelihood (see Appendix A.2). However, it is not projectively consistent over the actors (see Appendix A.3).

2.3 Observation model with noise

Allowing noise means that the model might not always realize an l from $\mathcal{L}[h]$. In this section, we explore a model that allows for such noise.

2.3.1 Repeated selection models

A draw $y \sim p_S(\cdot \mid h)$ from the noise-free model above can be realized sequentially, building up the actors in the list one at a time, just as the FM likelihood was motivated. These types

of models are classified as *repeated selection* models. Many other ranking models, including the Plackett–Luce model and the models mentioned above, can also be expressed as repeated selection models.

Definition 1 (Repeated selection model). A ranking model is called a *repeated selection model* if its likelihood can be written as

$$p(y \mid h) = \prod_{i=1}^n q_{y_{i:n}}(y_i \mid h[y_{i:n}]),$$

where

$$q_{y_{i:n}}(y_i \mid h[y_{i:n}])$$

is the probability of selecting y_i as the next element, conditional on the set of remaining actors $y_{i:n}$ and their induced suborder $h[y_{i:n}]$.

For instance, writing

$$\mathcal{L}_j[h] := \{\ell \in \mathcal{L}[h] : \ell_1 = j\}, \quad \forall j \in M,$$

as the set of all linear extensions of h whose first element is j , we can express the uniform likelihood model as a telescoping product over suborders:

$$p(y \mid h) = \prod_{i=1}^n \frac{|\mathcal{L}_{y_i}[h[y_{i:n}]]|}{|\mathcal{L}[h[y_{i:n}]]|}.$$

Notice that

$$\frac{|\mathcal{L}_{y_i}[h[y_{i:n}]]|}{|\mathcal{L}[h[y_{i:n}]]|}$$

gives the probability that y_i is selected as the next element in the list from the remaining actors under the uniform linear-extension model.

2.3.2 Queue jumping noise

Nicholls et al. [2025] allow for noise in the model by allowing actors in the partial order to “jump the queue” at every step in a *repeated selection* model. Let $i \leq n$. The model suggests that when we are deciding which element to put in the i ’th entry of the list, with probability $\epsilon \in [0, 1]$ we pick the next entry uniformly at random from the remaining actors (that is, with probability $\frac{1}{n-i+1}$ each), and with probability $1 - \epsilon$ we select the next entry according to the original repeated-selection model. Using the notation in the definition of a *repeated selection* model, we have

$$p(y \mid h, \epsilon) = \prod_{i=1}^n q_{y_{i:n}}(y_i \mid h[y_{i:n}], \epsilon),$$

where

$$q_{y_{i:n}}(y_i \mid h[y_{i:n}], \epsilon) = \frac{\epsilon}{n-i+1} + (1-\epsilon)q_{y_{i:n}}(y_i \mid h[y_{i:n}]).$$

Notice that when $\epsilon = 0$ the model reduces to the original *repeated selection* model, and when $\epsilon = 1$ it has the same distribution as $\mathcal{U}[\mathcal{P}_M]$.

One issue with the queue-jumping noise model is that it does not take into account the number of actors an actor skips when it is lifted up in the queue. A solution is to use the η -weighted queue-jumping model suggested by Jiang and Nicholls [2026]. However, we shall not consider it further for simplicity in this report.

3 Bayesian Inference for Partial Order with Tied Ranking Data

In this section, we will carry on with the setup as in Section 2.1, except that the data we are given are rankings with ties. We will model the data using the pair $(o, b) = (o, (b_1 \succ b_2 \succ \dots \succ b_N))$, where $o \subset M$ is a set with at least 2 elements (we choose 2 since it is the least amount of element in the ground set of which we can assign meaningful preference to). For each ranking, the first bucket $b_1 \subset o$ contains the most preferred choices, the second bucket $b_2 \subset o$ contains the second most preferred choices, and so on. For the purpose of exposition, we first consider the case where there is only one bucket-order observation. This can, however, be easily extended to the general case towards the end of the section.

3.1 Observation model

One way to generate a bucket-order observation from a true underlying partial order h_0 is through the procedure described in Algorithm 1.

Algorithm 1. *Observing a bucket order from a partial order.*

- (i) Sample $o \sim \rho(\cdot)$, where ρ is some distribution on the subsets of M with at least 2 elements.
- (ii) Observe a total order $y \sim p(\cdot \mid h_0[o])$.
- (iii) Observe a bucket order $b \sim g(\cdot \mid h_0[o], y)$, where g is some distribution over bucket orders.
- (iv) Record (o, b) .

In words, when observing a bucket order, the assessor first selects a subset $o \subseteq M$ according to some distribution ρ , where o is the set of elements from the ground set to be compared. Conditional on the selected subset o , the assessor then observes a total order $y \in \mathcal{P}_o$ according to the distribution $p(\cdot \mid h[o])$. Finally, the total order y is recorded as a bucket order b according to the distribution $g(\cdot \mid h, y, o)$. Let $\pi(\cdot)$ be the prior distribution of the underlying partial order h . Given an observed subset o and bucket order b , the posterior distribution of h is obtained by summing over all possible total orders $y \in \mathcal{P}_o$ that could have generated b :

$$\begin{aligned}
 p(h \mid b, o) &\propto \pi(h) p(b, o \mid h) \\
 &\propto \pi(h) \sum_{y \in \mathcal{P}_o} \mathbb{P}[\text{observing } y \text{ and recording } (b, o) \mid h] \\
 &\propto \pi(h) \sum_{y \in \mathcal{P}_o} \mathbb{P}[\text{recording } b \mid h, y, o] \mathbb{P}[\text{observing } y \text{ and } o \mid h] \\
 &\propto \pi(h) \sum_{y \in \mathcal{P}_o} g(b \mid h, y, o) p(y \mid h[o]) \rho(o \mid h).
 \end{aligned} \tag{1}$$

3.2 Ignorability

The posterior can still be complicated depending on the choice of g . To simplify the posterior to achieve our goal of inferring h_0 , we introduce two assumptions on g for the to simplify posterior. We say that the bucket-observation mechanism g is *ignorable* if, conditional on the observed subset $o \subseteq M$, it satisfies both of the following assumptions.

- (A1)** $g(b \mid h, y, o) = g(b \mid y, o)$ for all $h \in \mathcal{H}_M$, $o \subseteq M$, and $y \in \mathcal{P}_o$. That is, conditional on the observed subset o and the latent total order y , the distribution of b under g is

independent of the underlying partial order h .

(A2) For every $o \subseteq M$ and $y \in \mathcal{P}_o$,

$$g(b \mid y, o) \propto \mathbb{I}_{\{y \in \mathcal{L}[b]\}}.$$

That is, $g(b \mid y, o) = 0$ whenever $y \notin \mathcal{L}[b]$, and $g(b \mid y, o)$ takes the same value for all $y \in \mathcal{L}[b]$. We denote this common non-zero value by $c_{b,o}$.

Supposing that the subset-selection mechanism is independent of the underlying partial order and that g is ignorable, it then follows that

$$p(h \mid b, o) \propto \pi(h) \sum_{y \in \mathcal{P}_o} p(y \mid h[o]) g(b \mid h, y, o).$$

By assumption **(A1)**, conditional on the observed subset o and the latent total order y , the distribution of the observed bucket order b is independent of the underlying partial order h . Hence,

$$p(h \mid b, o) \propto \pi(h) \sum_{y \in \mathcal{P}_o} p(y \mid h[o]) g(b \mid y, o).$$

By assumption **(A2)**, $g(b \mid y, o)$ takes the same value for every $y \in \mathcal{L}[b]$ and is zero whenever $y \notin \mathcal{L}[b]$. Denoting this common non-zero value by c_b , we may write

$$g(b \mid y, o) = c_b \mathbb{I}_{\{y \in \mathcal{L}[b]\}}.$$

Therefore,

$$\begin{aligned} p(h \mid b, o) &\propto \pi(h) \sum_{y \in \mathcal{P}_o} p(y \mid h[o]) c_b \mathbb{I}_{\{y \in \mathcal{L}[b]\}} \\ &= \pi(h) c_b \sum_{y \in \mathcal{L}[b]} p(y \mid h[o]). \end{aligned}$$

Finally, c_b depends only on the observed bucket order b and not on h . Since b is fixed when considering the posterior as a function of h , c_b can be absorbed into the proportionality constant. Hence,

$$p(h \mid b, o) \propto \pi(h) \sum_{y \in \mathcal{L}[b]} p(y \mid h[o]). \quad (2)$$

This means that the choice of g can be neglected as long as it is *ignorable*, as the name suggests. But what are the assumptions really saying? One can think of **(A1)** as saying that the generation of b does not depend on the underlying partial order h once y is given, while **(A2)** tells us that (i) the relations in b cannot contradict those in y , and (ii) $g(b \mid y)$ does not favor any particular y among those that agree with b .

Here is an instance where g is ignorable. Suppose $o = M$ with probability 1 and we observe a linear extension y of h from a likelihood model. We then form a bucket order by writing the actors in y from top to bottom, moving down to a new bucket with probability $\tilde{p} \in [0, 1]$ independently at each step, as illustrated in Figure 3. Equivalently, each pair of consecutive actors in y is placed in the same bucket, or “collapsed”, with probability $1 - \tilde{p}$. Since a given bucket order b uniquely determines which of the $n - 1$ consecutive pairs in y must start a new bucket and which must collapse, g is then given by

$$g(b \mid h, y, o, \tilde{p}) = (\tilde{p})^{d(b)-1} (1 - \tilde{p})^{n-1} \mathbb{I}_{[y \in \mathcal{L}[b]]}, \quad (3)$$

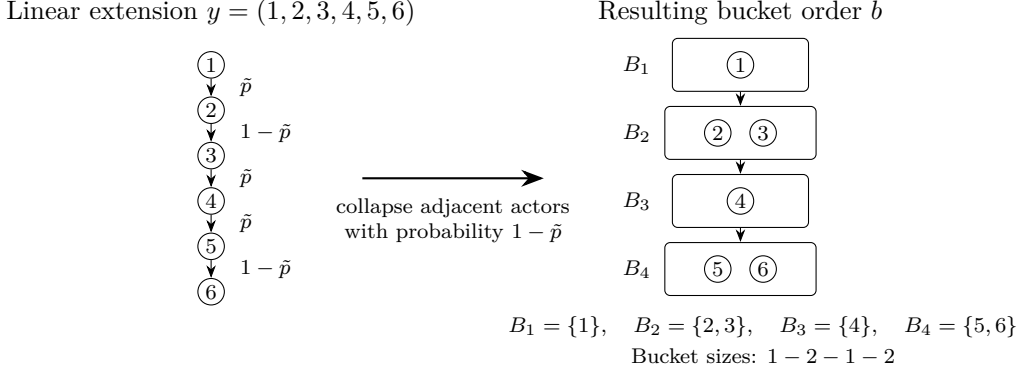


Figure 3: Illustration of the bucket-generation mechanism. Starting from the linear extension $y = (1, 2, 3, 4, 5, 6)$, adjacent actors are placed in the same bucket with probability $1 - \tilde{p}$ and a new bucket is started with probability $\tilde{p}$. In this realization, the edges between 2 and 3 and between 5 and 6 collapse, producing a bucket order with bucket sizes $1 - 2 - 1 - 2$.

where $d(b)$ is the depth of b . g clearly satisfies **(A1)** and **(A2)** since it is independent of h , vanishes whenever $y \notin \mathcal{L}[b]$, and is constant across all $y \in \mathcal{L}[b]$. This gives a glimpse to when g could be ignorable.

Claim 3.1. Fix $o \subseteq M$ and, WLOG, let $o = [|o|]$. For each $y \in \mathcal{P}_o$, let σ_y be the permutation such that $\sigma_y(y) = 1 \succ 2 \succ \dots \succ |o|$. For each $o \subset M$ where $|o| \geq 2$, let D_o be a random variable supported on

$$\{b \in \mathcal{B}_o : b \text{ agrees with } 1 \succ 2 \succ \dots \succ |o|\},$$

and define

$$g(b \mid y, o) = \mathbb{P}[D_o = \sigma_y(b)].$$

Then g is ignorable.

Proof. Since g does not depend on h , **(A1)** holds. It remains to verify **(A2)**.

If $y \notin \mathcal{L}[b]$, then $\sigma_y(y) \notin \mathcal{L}[\sigma_y(b)]$. Since $\sigma_y(y) = 1 \succ \dots \succ |o|$, $\sigma_y(b)$ lies outside the support of D_o , and hence $\mathbb{P}[D_o = \sigma_y(b)] = 0$ and $g(b \mid y, o) = 0$.

Now let $y, y' \in \mathcal{L}[b]$. Since they can differ only by permutations within the buckets of b , there exists a permutation α such that $\alpha(y) = y'$ and $\alpha(b) = b$. Hence $\alpha^{-1}(b) = b$, and

$$\begin{aligned}
 g(b \mid y, o) &= \mathbb{P}[D_o = \sigma_y(b)] \\
 &= \mathbb{P}[D_o = (\sigma_y \circ \alpha^{-1})(b)] \\
 &= g(b \mid (\sigma_y \circ \alpha^{-1})^{-1}(1 \succ \dots \succ |o|), o) \\
 &= g(b \mid \alpha(y), o) = g(b \mid y', o).
 \end{aligned}$$

Thus $g(b \mid y, o)$ is constant over $y \in \mathcal{L}[b]$ and zero otherwise, so

$$g(b \mid y, o) \propto \mathbb{I}_{\{y \in \mathcal{L}[b]\}}.$$

Therefore **(A2)** holds, and g is ignorable. $\square$

Heuristically Claim 3.1 is saying g is ignorable when it is only a weight on the structure of the bucket. It's also obvious the ignorable instance of g we just gave also belongs to this family of distribution. However, the converse is not necessarily true - not all ignorable g can be characterized with a family of random variables $\{D_o\}_{o \subseteq M}$. See Appendix A.4 for a counterexample.

3.3 MCMC targeting the posterior

The main purpose of introducing ignorability is to simplify the posterior when we have multiple observations

$$\mathcal{D}_N = \{(o^{(i)}, b^{(i)})\}_{i=1}^N.$$

Suppose that the subset-selection mechanism ρ is independent of h , that the bucket-observation mechanism g_i associated with each observation is ignorable, and that the observations are conditionally independent given h . Then

$$\begin{aligned} p(h \mid \mathcal{D}_N) &\propto \pi(h) \prod_{i=1}^N \sum_{y \in \mathcal{P}_{o^{(i)}}} p(y \mid h[o^{(i)}]) g_i(b^{(i)} \mid h, y, o^{(i)}) \\ &\propto \pi(h) \prod_{i=1}^N \sum_{y \in \mathcal{P}_{o^{(i)}}} p(y \mid h[o^{(i)}]) c_{b^{(i)}} \mathbb{I}_{\{y \in \mathcal{L}[b^{(i)}]\}} \\ &\propto \pi(h) \prod_{i=1}^N \sum_{y \in \mathcal{L}[b^{(i)}]} p(y \mid h[o^{(i)}]). \end{aligned} \tag{4}$$

Here, the factors $\rho(o^{(i)})$ and $c_{b^{(i)}}$ have been absorbed into the proportionality constant as neither depends on h . It's also worth noting that under ignorability the bucket-observation mechanisms g_i do not enter the posterior, so inference for h only requires the observed subsets $o^{(i)}$, the corresponding bucket orders $b^{(i)}$, and the ranking likelihood $p(y \mid h[o^{(i)}])$.

Notice as every partial order $h \in \mathcal{H}_M$ can be represented by a directed acyclic graph (DAG) $G = (M, E)$, where the vertices are the actors and the directed edges represent the ordering relations, to simulate from the posterior in (4) we run a pseudo-marginal MCMC targeting the DAGs representing h using auxiliary total orders y , as described in Algorithm 2. The partial order h corresponding to G is obtained by taking its transitive closure, which we denote by $C(G)$.

There are generally several DAGs whose transitive closure gives the same partial order. Let $R(G)$ denote the transitive reduction of G and define

$$w(G) = |E(C(G))| - |E(R(G))|.$$

The relations in the transitive closure that are not present in the transitive reduction are implied by transitivity. Each such relation can therefore either be included or omitted in the DAG without changing the underlying partial order. Hence, a partial order h corresponds to $2^{w(h)}$ different DAGs.

Under this DAG representation of h , the prior distribution is no longer simply $\pi(h)$, since a single partial order can be represented by a multiple ($2^{w(h)}$) DAGs. We therefore need to account for this factor when defining the prior on the augmented DAG space. Specifically for a DAG G , we take

$$\tilde{\pi}(G) = \frac{\pi(C(G))}{2^{w(C(G))}}.$$

The factor $2^{-w(C(G))}$ corrects for the number of DAG representations of the same partial order. Indeed, summing the prior probabilities over all DAGs representing a given partial order h gives

$$\sum_{G: C(G)=h} \tilde{\pi}(G) = 2^{w(h)} \frac{\pi(h)}{2^{w(h)}} = \pi(h).$$

Thus, although the MCMC is run on the space of DAGs, the induced prior on their transitive closures remains the original prior π on partial orders.

We now construct an MCMC algorithm whose states are DAGs. At each update, an ordered pair of distinct vertices is selected uniformly at random. If the corresponding directed

edge is present, we propose removing it; otherwise, we propose adding it. Proposals that introduce a directed cycle are rejected. For a valid proposal G' from the current DAG G , the Metropolis–Hastings acceptance probability is

$$\alpha_G = 1 \wedge \frac{\tilde{\pi}(G') \prod_{i=1}^N p(y^{(i)} | C(G')[o^{(i)}])}{\tilde{\pi}(G) \prod_{i=1}^N p(y^{(i)} | C(G)[o^{(i)}])} \frac{q(G | G')}{q(G' | G)}.$$

For the symmetric edge-addition and edge-deletion proposal used here, the proposal ratio is one. Under the uniform prior on partial orders, the acceptance probability therefore becomes

$$\alpha_G = 1 \wedge \left[2^{w(C(G)) - w(C(G'))} \prod_{i=1}^N \frac{p(y^{(i)} | C(G')[o^{(i)}])}{p(y^{(i)} | C(G)[o^{(i)}])} \right] \mathbb{I}_{\{G' \text{ is a DAG}\}}. \quad (5)$$

The auxiliary total orders $y^{(i)} \in \mathcal{L}[b^{(i)}]$ allow the posterior likelihood to be evaluated without explicitly summing over all linear extensions. The MCMC state therefore consists of both the DAG G and the collection of auxiliary total orders $y = (y^{(1)}, \dots, y^{(N)})$.

We provide an outline of the exact algorithm we used in the simulation in Algorithm 2, but exact implementation and the code used for the experiments is available on GitHub. Despite not included in Algorithm 2, we use the FM likelihood together with an ϵ -queue jumping model as our choice of the observation model $p(y | h[o])$, also as mentioned we are assuming the bucket generating process g_i are ignorable.

Algorithm 2. *MCMC targeting the posterior in (4).*

Input: $\mathcal{D}_N = \{(o^{(i)}, b^{(i)})\}_{i=1}^N$ and number of iterations T .

Initialize: Choose a DAG G_0 and, for each $i = 1, \dots, N$, choose

$$y^{(i,0)} \in \mathcal{L}[b^{(i)}].$$

For $t = 0, \dots, T - 1$:

With probability r , perform a G -update:

(i) Choose a pair of distinct vertices (j_1, j_2) uniformly at random.

(ii) If $(j_1, j_2) \in E_t$, propose

$$E' = E_t \setminus \{(j_1, j_2)\};$$

otherwise propose

$$E' = E_t \cup \{(j_1, j_2)\}.$$

(iii) Let $G' = (M, E')$. If G' contains a directed cycle, reject the proposal. Otherwise compute its transitive closure $C(G')$.

(iv) Accept G' with probability

$$\alpha_G = 1 \wedge \left[2^{w(C(G_t)) - w(C(G'))} \prod_{i=1}^N \frac{p(y^{(i,t)} | C(G')[o^{(i)}])}{p(y^{(i,t)} | C(G_t)[o^{(i)}])} \right].$$

Otherwise set $G_{t+1} = G_t$.

With probability $1 - r$, perform a y -update:

(i) Choose

$$j \sim \text{Unif}\{1, \dots, N\}.$$

(ii) Sample

$$y' \sim \text{Unif}(\mathcal{L}[b^{(j)}]).$$

(iii) Accept y' with probability

$$\alpha_y = 1 \wedge \frac{p(y' \mid C(G_t)[o^{(j)}])}{p(y^{(j,t)} \mid C(G_t)[o^{(j)}])}.$$

Otherwise set $y^{(j,t+1)} = y^{(j,t)}$.

(iv) For all $i \neq j$, set

$$y^{(i,t+1)} = y^{(i,t)}.$$

Return: the sampled partial orders

$$C(G_0), C(G_1), \dots, C(G_T).$$

3.4 Asymptotic consistency

We recall some notation for this section. Let $h_0 \in \mathcal{H}_M$ denote the true underlying partial order, and let

$$\mathcal{D}_N = \{X_i\}_{i=1}^N, \quad X_i = (o^{(i)}, b^{(i)}),$$

where the X_i are independently and identically distributed according to P_{h_0} . The sample space is

$$\mathcal{X} = \bigcup_{\substack{o \subseteq M \\ |o| \geq 2}} (\{o\} \times \mathcal{B}_o).$$

For each $h \in \mathcal{H}_M$, let P_h denote the distribution of a single observation under h , and write the posterior after N observations as

$$\Pi_N(h) = \mathbb{P}(h \mid \mathcal{D}_N) \propto \pi(h) \prod_{i=1}^N P_h(X_i).$$

We assume that the model is identifiable, in the sense that

$$h \neq h' \implies P_h \neq P_{h'}, \quad h, h' \in \mathcal{H}_M.$$

Theorem 1 (Posterior consistency). Suppose the model is identifiable and $\pi(h) > 0$ for every $h \in \mathcal{H}_M$. Then, for every $h \in \mathcal{H}_M$,

$$\Pi_N(h) \xrightarrow{\text{a.s.}} \mathbb{I}_{\{h=h_0\}} \quad \text{as } N \rightarrow \infty.$$

Proof. Fix $h \neq h_0$. The posterior odds of h against h_0 satisfy

$$\frac{1}{N} \log \frac{\Pi_N(h)}{\Pi_N(h_0)} = \frac{1}{N} \log \frac{\pi(h)}{\pi(h_0)} + \frac{1}{N} \sum_{i=1}^N \log \frac{P_h(X_i)}{P_{h_0}(X_i)}. \quad (6)$$

Since $\pi(h), \pi(h_0) > 0$, the first term converges to zero. By the strong law of large numbers,

$$\frac{1}{N} \sum_{i=1}^N \log \frac{P_h(X_i)}{P_{h_0}(X_i)} \xrightarrow{\text{a.s.}} -D_{\text{KL}}(P_{h_0} \| P_h).$$

Identifiability implies $P_h \neq P_{h_0}$ and hence

$$c_h := D_{\text{KL}}(P_{h_0} \| P_h) > 0.$$

Therefore

$$\log \frac{\Pi_N(h)}{\Pi_N(h_0)} = -Nc_h + o(N) \quad \text{a.s.},$$

and consequently

$$\frac{\Pi_N(h)}{\Pi_N(h_0)} = \exp\{-Nc_h + o(N)\} \xrightarrow{\text{a.s.}} 0.$$

Since $\Pi_N(h_0) \leq 1$, this implies $\Pi_N(h) \rightarrow 0$ almost surely for every $h \neq h_0$.

Finally, $\mathcal{H}_M$ is finite, so

$$\Pi_N(h_0) = 1 - \sum_{h \neq h_0} \Pi_N(h) \xrightarrow{\text{a.s.}} 1.$$

□

An immediate consequence of Theorem 1 is the consistency of the maximum a posteriori estimator

$$\hat{h}_N^{\text{MAP}} \in \arg \max_{h \in \mathcal{H}_M} \Pi_N(h).$$

Indeed, since $\Pi_N(h_0) \rightarrow 1$ almost surely while $\Pi_N(h) \rightarrow 0$ for every $h \neq h_0$,

$$\hat{h}_N^{\text{MAP}} = h_0$$

eventually almost surely.

While the MAP is an asymptotically consistent estimator for h_0 , finding it might be computationally difficult. We therefore consider a family of estimators based on posterior probabilities of individual order relations.

Definition 2 (q -consensus). For $q \in [0, 1]$, define the binary matrix $M_q^{(N)}$ by

$$(M_q^{(N)})_{ij} = \mathbb{I} \{ \Pi_N(i \succ_h j) \geq q \},$$

where

$$\Pi_N(i \succ_h j) := \sum_{h \in \mathcal{H}_M} \mathbb{I}_{\{i \succ_h j\}} \Pi_N(h).$$

The q -consensus matrix $Q_q^{(N)}$ is the transitive closure of $M_q^{(N)}$. Whenever $Q_q^{(N)}$ represents a partial order, we call that partial order the q -consensus order.

Corollary 2 (q -consensus consistency). Under the assumptions of Theorem 1, for every fixed $q \in (0, 1)$,

$$Q_q^{(N)} = h_0$$

eventually almost surely.

Proof. By Theorem 1,

$$\Pi_N(i \succ_h j) \xrightarrow{\text{a.s.}} \mathbb{I}_{\{i \succ_{h_0} j\}}$$

for every $i, j \in M$. Since $q \in (0, 1)$, it follows that

$$(M_q^{(N)})_{ij} \xrightarrow{\text{a.s.}} \mathbb{I}_{\{i \succ_{h_0} j\}}.$$

There are finitely many pairs (i, j) , so $M_q^{(N)}$ eventually equals the adjacency matrix of h_0 almost surely. Since h_0 is already transitively closed, its transitive closure is itself, and hence $Q_q^{(N)} = h_0$ eventually almost surely. □

We finally verify the identifiability condition of Theorem 1 for the FM likelihood with queue-jumping noise.

Corollary 3. Suppose the ranking likelihood is the FM model with queue-jumping probability $\epsilon \in [0, 1)$, and assume:

1. every two-element subset has positive probability of being selected, i.e.

$$\rho(\{i, j\}) > 0 \quad \forall i \neq j;$$

2. conditional on $o = \{i, j\}$, the bucket-observation mechanism record the two actors in separate buckets with non-zero probability.

Then the model is identifiable. As a result of Theorem 1, if the prior has full support on $\mathcal{H}_M$ then the posterior is asymptotically consistent.

Proof. Take $h \neq h'$. Then there exist $i, j \in M$ whose relation differs between h and h' . Without loss of generality, suppose $i \succ_h j$ but $i \not\succ_{h'} j$, and take $o = \{i, j\}$.

Under $h[o]$, the probability that the latent total order is $i \succ j$ is

$$p(i \succ j \mid h[o]) = 1 - \frac{\epsilon}{2}.$$

Under $h'[o]$, we instead have

$$p(i \succ j \mid h'[o]) = \begin{cases} \frac{1}{2}, & \text{if } i \text{ and } j \text{ are incomparable under } h'[o], \\ \frac{\epsilon}{2}, & \text{if } j \succ_{h'} i. \end{cases}$$

These three probabilities are distinct whenever $\epsilon < 1$.

Since $\rho(\{i, j\}) > 0$ and the bucket-observation mechanism has positive probability of recording the strict order $i \succ j$, there is an observed event

$$X = (\{i, j\}, i \succ j)$$

whose probability differs under h and h' . Hence $P_h \neq P_{h'}$. The reverse direction is immediate - if $h = h'$, then $P_h = P_{h'}$. Thus the model is identifiable, and the result follows from Theorem 1. $\square$

Now suppose $X_1, \dots, X_N$ are observed. For $i, j \in M$, define

$$R_{i,j}^{(N)} := \mathbb{P}(i \succ_h j \mid h \sim \Pi_N) = \sum_{h \in \mathcal{H}_M} \mathbb{I}_{\{i \succ_h j\}} \Pi_N(h),$$

the posterior probability that $i \succ j$. Also let

$$\epsilon_N := 1 - \Pi_N(h_0)$$

denote the posterior mass assigned to all partial orders other than h_0 .

If $i \succ_{h_0} j$, then the term corresponding to h_0 appears in the sum defining $R_{i,j}^{(N)}$, and hence

$$R_{i,j}^{(N)} \geq \Pi_N(h_0) = 1 - \epsilon_N. \quad (7)$$

On the other hand, if $i \not\succ_{h_0} j$, then every partial order containing the relation $i \succ j$ must differ from h_0 . Therefore,

$$R_{i,j}^{(N)} \leq \sum_{h \neq h_0} \Pi_N(h) = \epsilon_N. \quad (8)$$

Recall that the q -consensus correctly classifies the relation $i \succ j$ whenever

$$\begin{cases} R_{i,j}^{(N)} \geq q, & \text{if } i \succ_{h_0} j, \\ R_{i,j}^{(N)} < q, & \text{if } i \not\succ_{h_0} j. \end{cases}$$

Using (7) and (8), it is therefore sufficient that

$$\begin{cases} 1 - \epsilon_N \geq q, \\ \epsilon_N < q. \end{cases}$$

Equivalently,

$$\begin{cases} \epsilon_N \leq 1 - q, \\ \epsilon_N < q. \end{cases}$$

Thus the stronger condition

$$\epsilon_N < \min(q, 1 - q) =: E_q \tag{9}$$

is sufficient to guarantee

$$Q_q^{(N)} = h_0.$$

By Theorem 1, $\epsilon_N \rightarrow 0$ almost surely as $N \rightarrow \infty$. Hence, for every fixed $q \in (0, 1)$, exact recovery is eventually achieved once $\epsilon_N < E_q$.

This also provides a reason to choose $q = 1/2$ in practice. Since

$$E_q = \min(q, 1 - q)$$

is maximized at $q = 1/2$, this choice requires the least posterior concentration for the condition (9) to hold.

3.4.1 Finite-sample convergence rate

We now derive a finite-sample bound for the probability that the q -consensus differs from h_0 . Assume that

$$\pi(h) > 0 \quad \forall h \in \mathcal{H}_M,$$

and that

$$P_h(x) > 0 \quad \forall h \in \mathcal{H}_M, \quad x \in \mathcal{X}.$$

For $X \sim P_{h_0}$ and $h \in \mathcal{H}_M$, define the log-likelihood ratio

$$Z(h) := \log \left(\frac{P_h(X)}{P_{h_0}(X)} \right).$$

For each $h \neq h_0$, let

$$c_h := D_{\text{KL}}(P_{h_0} \parallel P_h),$$

so that

$$\mathbb{E}_{h_0}[Z(h)] = -c_h.$$

We further define

$$\Delta := \min_{h \neq h_0} D_{\text{KL}}(P_{h_0} \parallel P_h),$$

which is strictly positive by identifiability, and

$$R := \max_{h \neq h_0} \left\{ \max_{x \in \mathcal{X}} \log \frac{P_h(x)}{P_{h_0}(x)} - \min_{x \in \mathcal{X}} \log \frac{P_h(x)}{P_{h_0}(x)} \right\}.$$

Since $\mathcal{X}$ is finite and all probabilities are positive, $R < \infty$.

Finally, define

$$A := \sum_{h \neq h_0} \frac{\pi(h)}{\pi(h_0)} = \frac{1 - \pi(h_0)}{\pi(h_0)}.$$

The following lemma gives an exponential upper bound on the probability that the q -consensus fails to recover h_0 .

Lemma 4. Suppose $\pi(h) > 0$ for every $h \in \mathcal{H}_M$ and $P_h(x) > 0$ for every $h \in \mathcal{H}_M$ and $x \in \mathcal{X}$. For any $q \in (0, 1)$, if

$$N > \frac{2}{\Delta} \log \left(\frac{A}{E_q} \right),$$

then

$$\mathbb{P}_{h_0}(Q_q^{(N)} \neq h_0) \leq (|\mathcal{H}_M| - 1) \exp \left(-\frac{\Delta^2}{2R^2} N \right).$$

Proof. See Appendix A.6. □

The convergence of the expected the Hamming error now follows as a corollary of the lemma.

Theorem 5 (Expected Hamming error). Under the assumptions of Lemma 4, if

$$N > \frac{2}{\Delta} \log \left(\frac{A}{E_q} \right),$$

then

$$\mathbb{E}_{h_0} [d_H(Q_q^{(N)}, h_0)] \leq |M|(|M| - 1) (|\mathcal{H}_M| - 1) \exp \left(-\frac{\Delta^2}{2R^2} N \right),$$

where d_H denotes the Hamming distance between the relation matrices of the two partial orders.

Proof. There are at most $|M|(|M| - 1)$ directed relations between distinct actors. Hence

$$d_H(Q_q^{(N)}, h_0) \leq |M|(|M| - 1) \mathbb{I}_{\{Q_q^{(N)} \neq h_0\}}.$$

Taking expectations gives

$$\mathbb{E}_{h_0} [d_H(Q_q^{(N)}, h_0)] \leq |M|(|M| - 1) \mathbb{P}_{h_0}(Q_q^{(N)} \neq h_0).$$

Applying Lemma 4 yields

$$\mathbb{E}_{h_0} [d_H(Q_q^{(N)}, h_0)] \leq |M|(|M| - 1) (|\mathcal{H}_M| - 1) \exp \left(-\frac{\Delta^2}{2R^2} N \right),$$

as required. □

4 Simulation Experiments

To demonstrate the effectiveness of the MCMC introduced in Algorithm 2, we tested the chain under the uniform and uniform-depth priors using the FM model in several different cases. In every case in this section, we use m for the number of actors in the ground set and N for the number of lists/sublists given as data. We also set $r = .5$ (as in Algorithm 2) for each test case. All CIs are reported at the 95% confidence level.

4.1 Data generation

We generate the data as follows. Suppose we are given the true partial order h_0 on ground set $M = \{1, \dots, m\}$ and are required to generate N bucket orders $\{(o_i, b_i)\}_{i=1}^N$. For each $i = 1, \dots, N$ we first sample n_i , the number of actors in bucket order b_i uniformly at random from $\{2, 3, \dots, m\}$, then we sample n_i elements uniformly at random without replacement from M to create o_i . Once o_i has been generated we then generate a total order l_i according to the FM observation model with or without queue-jumping error. We then create the bucket order b_i according to the bucket-generation model described in Figure 3, with $\tilde{p} = 0.5$. The exact implementation and the code is available on GitHub.

4.2 Chain Analysis

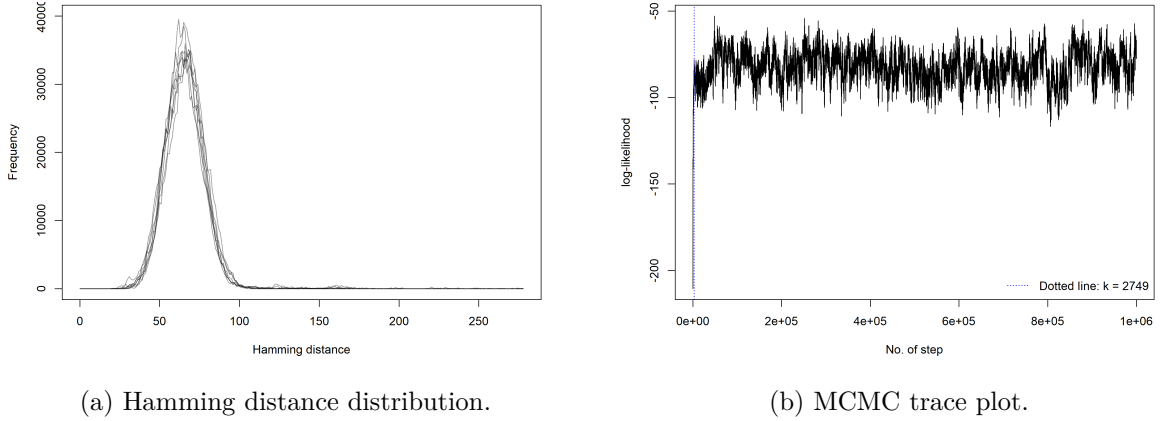


Figure 4: Simulation results and MCMC diagnostics.

To investigate the samples of the chain, we ran 8 chains with 10^6 proposals using $N = 10$ suborder bucket-order observations with noise ($\epsilon = .05$), generated from the same true poset of $m = 20$ actors, and initialized the chains at different posets.

The log-likelihood trace plots show that the chains take around 5 thousand steps to reach equilibrium. However, the chains are also highly autocorrelated. The effective sample size (ESS) was calculated for each non-constant entry of the adjacency matrices using samples from the $10^5 + 1$ th to the 10^6 th steps of each of the 8 chains. The minimum ESS across these entries is around 96.25 ± 11.27 , meaning that it takes around 9350 steps per effective sample on average. The average acceptance ratio across the 8 chains is $.1685 \pm .0173$, of which $.1535 \pm .0007$ results from accepting an h -update.

On our computer, the chains took on average around $.7354 \pm .0052$ hours to run 10^6 steps, giving one effective sample roughly every 25.22 ± 3.18 seconds after equilibrium. A table showing the runtime of the algorithm for different values of n and N is also provided in Appendix B.1.

While it is not possible to plot the full empirical posterior distribution over h , we can compare the empirical posterior distributions of the Hamming distance from the true poset across the chains in Figure 4, after discarding the first 10^5 samples as burn-in. As we can see, the distributions of the 8 chains are quite close to one another, providing evidence that the chains have converged to the same stationary distribution.

4.3 Predictive performance

The goal of the report has always been finding ways to retrieve the central partial order h_0 . We wish to estimate h_0 using a q -consensus partial order Q_q^N (if exist). However this would

not be possible since the posterior is by no means easy to compute. Instead, we estimate h_0 using a consistent estimator of Q_q^N , as defined below:

Definition 3 (Empirical q -consensus matrix). Let

$$h^{(1)}, \dots, h^{(K)}$$

be K partial orders sampled from the posterior distribution Π_N , and let $M^{(k)}$ denote the adjacency matrix of $h^{(k)}$. Define the empirical posterior relation probability matrix $\hat{P}^{(K)}$ by

$$\hat{P}_{ij}^{(K)} = \frac{1}{K} \sum_{k=1}^K M_{ij}^{(k)} = \frac{1}{K} \sum_{k=1}^K \mathbb{I}_{\{i \succ_{h^{(k)}} j\}}.$$

Thus, $\hat{P}_{ij}^{(K)}$ is the empirical estimate of the posterior relation probability

$$R_{ij}^{(N)} = \mathbb{P}(i \succ_h j \mid h \sim \Pi_N).$$

For $q \in [0, 1]$, define the empirical q -consensus matrix $\hat{Q}_q^{(N,K)}$ entrywise by

$$\left(\hat{Q}_q^{(N,K)}\right)_{ij} = \mathbb{I}\left\{\hat{P}_{ij}^{(K)} \geq q\right\}.$$

When $\hat{Q}_q^{(N,K)}$ represents a partial order, we call the corresponding partial order the *empirical q -consensus order*. When there is no ambiguity about the number of posterior samples K , we drop K from the notation and simply write $\hat{P}$ and $\hat{Q}_q^{(N)}$.

For fixed N , $\hat{P}_{ij}^{(K)}$ is a consistent estimator of $R_{ij}^{(N)}$. In particular, if the samples $h^{(1)}, \dots, h^{(K)}$ are obtained from an ergodic MCMC chain with stationary distribution Π_N , then the ergodic theorem gives

$$\hat{P}_{ij}^{(K)} \xrightarrow{\text{a.s.}} R_{ij}^{(N)} \quad \text{as } K \rightarrow \infty.$$

Therefore, provided

$$R_{ij}^{(N)} \neq q \quad \text{for all } i, j \in M,$$

the thresholded entries also converge almost surely,

$$\left(\hat{Q}_q^{(N,K)}\right)_{ij} \xrightarrow{\text{a.s.}} \left(Q_q^{(N)}\right)_{ij}.$$

Since M is finite, this holds simultaneously for all pairs, and hence

$$\hat{Q}_q^{(N,K)} \xrightarrow{\text{a.s.}} Q_q^{(N)} \quad \text{as } K \rightarrow \infty.$$

Thus, away from the boundary case $R_{ij}^{(N)} = q$, the empirical q -consensus matrix is a consistent estimator of the theoretical q -consensus matrix.

Proposition 6. Let $h_0 \in \mathcal{H}_M$, where $|M| = m$, and let $s(h_0)$ denote the number of relations in the transitive closure of h_0 . Then

$$\max_{h \in \mathcal{H}_M} d_H(h, h_0) = \binom{m}{2} + s(h_0).$$

Moreover, the maximum is attained by any linear extension of the dual partial order h_0^t .

Proof. There are $s(h_0)$ comparable unordered pairs in h_0 and $\binom{m}{2} - s(h_0)$ incomparable pairs. For a comparable pair, the maximum Hamming contribution is 2, achieved by reversing the relation, while for an incomparable pair the maximum contribution is 1. Thus

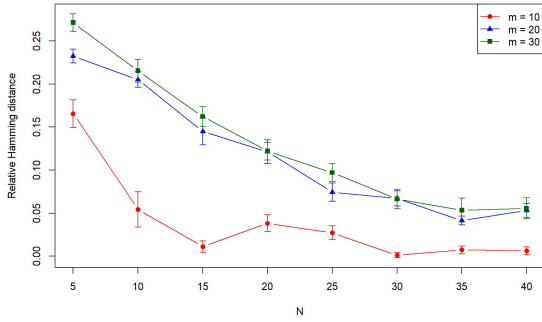
$$d_H(h, h_0) \leq 2s(h_0) + \binom{m}{2} - s(h_0) = \binom{m}{2} + s(h_0).$$

Let ℓ be any linear extension of the dual order h_0^t . Every relation in h_0 is reversed in ℓ , while every incomparable pair in h_0 is ordered in one direction by ℓ . Hence ℓ attains the bound. $\square$

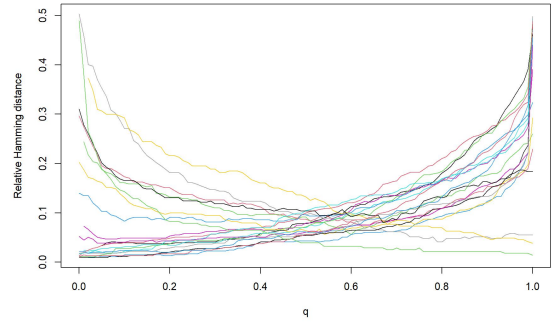
We call the *relative Hamming distance* of a partial order $h \in \mathcal{H}_M$ from h_0 to be

$$d_H^{\text{rel}}(h, h_0) := \frac{d_H(h, h_0)}{\binom{m}{2} + s(h_0)}.$$

The relative Hamming distance represents the proportion of the maximum possible Hamming distance from h_0 attained by h .



(a) Relative Hamming distance against N .



(b) Relative Hamming distance against q for $m = 20$.

Figure 5: Relative Hamming distance of the empirical q -consensus under different simulation settings.

The first thing we want to investigate is how efficient a $.5$ -consensus matrix $\hat{Q}_{.5}^{(N)}$ in predicting the true partial order h_0 among all other choices of $q \in (0, 1)$. To do this we ran the chain 20 times, each targeting a different partial order with $m = 24$. We have taken $N = 30$ and the queue jumping error $\epsilon = .05$, burned the first 10^5 samples and computed the empirical q -consensus matrix $\hat{Q}_q^{(N)}$ of the next 10^6 proposals (these numbers are chosen to match the parameter we use later in the primate data). Figure 5(b) shows how the relative Hamming distance between the q -consensus and the true partial order h_0 varies with q . Across all 20 simulation runs, the maximal of relative Hamming distance is minimized at a value of q close to 0.5. It is worth noting that nearly all of the q -consensus matrices we obtained represent a DAG, with a few exceptions when $q = 0$ to .02.

We then want to investigate whether if the q -consensus is consistent in inferring h_0 at extreme depth. Again we tested the algorithm at $m = 24$ when the true partial order h_0 is an empty order (all actors are unordered) or a total order ($1 \succ 2 \succ \dots \succ 24$) for 80 times. We have taken $N = 30$ and the queue jumping error $\epsilon = .05$. For each chain we burned the first 10^5 samples and found the empirical $.5$ -consensus $\hat{Q}_{.5}^{(N)}$ of the next 10^6 samples. When h_0 is the empty order, the $.5$ -consensus $\hat{Q}_{.5}^{(N)}$ has a relative Hamming distance of $.0166 \pm 0.0004$ from h_0 . When h_0 is a total order, the relative Hamming distance is 0.0293 ± 0.0023 , implying the $.5$ -consensus order (when exists) differs from the true partial order by approximately 1.66%

and 2.93% of the maximum possible Hamming distance in the empty and total-order cases, respectively. Figure 9 shows the .5-consensus partial order (when exists) obtain from 1 of the 80 chains in both case.

One might notice that the curves in Figure 9 tend to increase as q approaches either 0 or 1, although for some partial orders the increase is much more pronounced at only one of the two extremes; one possible explanation is the number of relations in the underlying partial order, since a smaller q may be preferable when h_0 contains many relations by retaining more posterior-supported relations in the q -consensus, whereas a larger q may be preferable when h_0 contains relatively few relations by avoiding the inclusion of relations that are not present in h_0 .

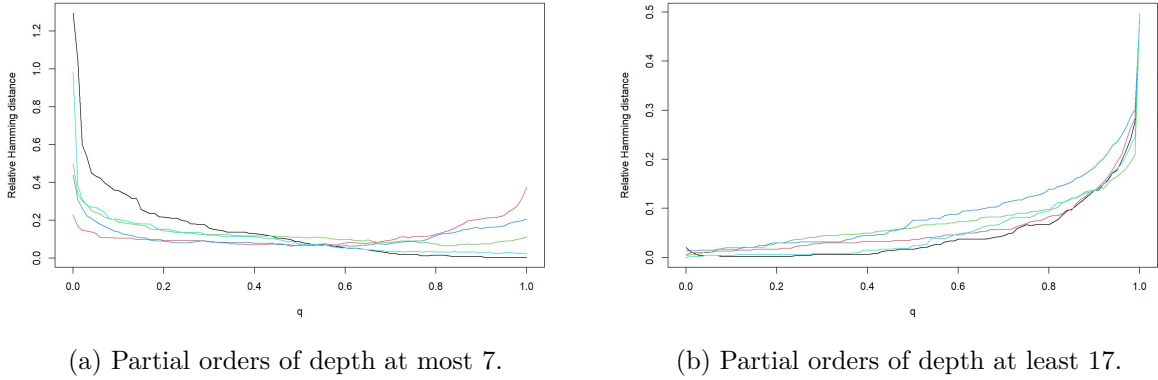


Figure 6: Relative Hamming distance of the empirical q -consensus against q for partial orders of different depths.

To investigate this effect, we ran 10 chains using the same setup as the experiment in Figure 9, with five chains targeting different partial orders of depth at most 7 and the remaining five targeting partial orders of depth at least 17; the mean numbers of relations in the lower- and higher-depth orders were 56 ± 83.3 and 269.4 ± 5.93 , respectively. The results are summarized in Figure 6: for lower-depth partial orders, the relative Hamming distance tends to remain smaller for larger values of q , whereas for higher-depth partial orders it tends to remain smaller for smaller values of q , which is consistent with the intuition that the preferred threshold may depend on the density of relations in the underlying partial order.

To investigate the reliability of $\hat{Q}_q^{(N)}$ as an estimator of h_0 as N increases, we applied Algorithm 2 at $m = 10, 20, 30$, generating 40 observations in each case and estimating the posterior using the first $N = 5, 10, 15, \dots, 40$ observations with queue-jumping error $\epsilon = .05$; we repeated each experiment 10 times to obtain 95% confidence intervals, discarding the first 10^5 samples of each chain as burn-in and computing the empirical .5-consensus from the subsequent 10^6 samples.

Figure 5(a) shows the relative Hamming distance of their .5-consensus matrix from their respective true partial order. The curve levels ceased to decrease when $m = 10$ at large N as most of the chains has predicted all edges correctly. The curves for $m = 20$ and $m = 30$ exhibit a roughly linear decrease in relative Hamming distance over the range of N considered; this does not contradict Theorem 5, which provides only an exponential upper bound on the expected Hamming distance, and a sufficiently slow exponential decay may also appear approximately linear over a limited range of N . The occasional increase in the mean relative Hamming distance despite an increase in N may be due to the presence of particularly noisy observations. (For instance the increase we see from $m = 10$ when N increases from 15 to 20)

4.4 Primate data

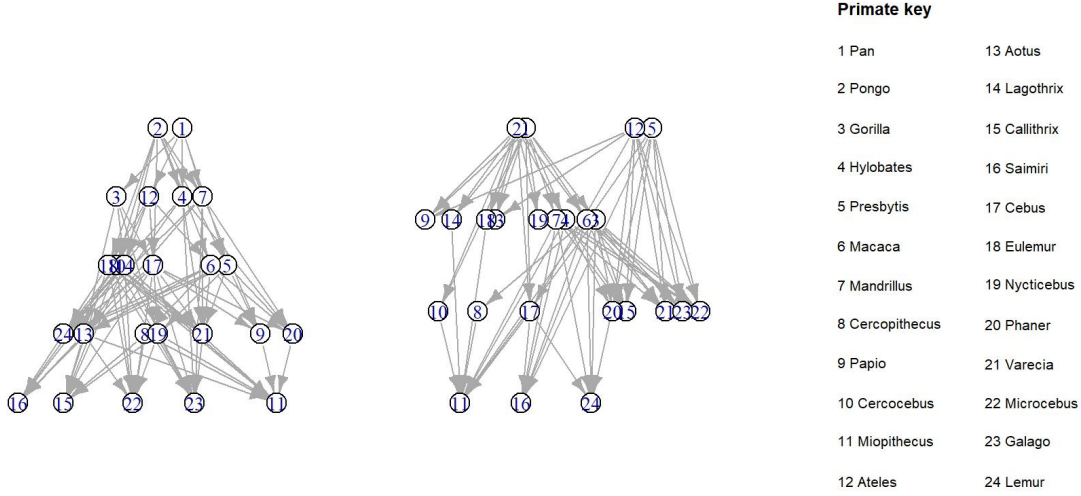


Figure 7: Consensus orders of the cognitive rankings of the 24 primate taxa. The partial order on the left corresponds to $q = 1/3$, while the partial order on the right corresponds to $q = 1/2$.

We now apply the MCMC to the primate rankings from different cognitive tasks obtained from Johnson et al. [2002]. We have $m = 24$ and $N = 30$, again we take $\epsilon = 0.05$ as the queue jumping error, and run the chains under the ignorability assumption. We ran 8 chains from different starting states, burned the first 10^5 samples and used the next 10^6 samples to obtain their .5-consensus orders. The maximum Hamming distance between the .5-consensus matrices of the 8 chains is 27 out of a probable maximum of $23 \times 24 = 552$, indicating that the 8 consensus orders are considerably similar. We have given the a .5-consensus order from one chain that has a minimal summed Hamming distance to the other 7 .5-consensus orders in Figure 7.

It is obvious that this .5-consensus is of depth 4. If we assume also that the underlying partial should have a greater depth, one can learn from our discussion about the relation between the true partial order’s depth and the q -consensus matrices from Section 4.3 and reduce q . Reducing q to $1/3$ gives us a partial order of depth 5 that is slightly more assertive about the relations and interesting to look at.

Both results from Figure 7 is quite similar to that obtained by Johnson et al. [2002], with *Pan* and *Pongo* ranked near the top, followed by *Ateles*, and *Miopithecus* near the bottom. We can also observe a general decreasing trend in absolute brain mass as one moves down the poset, suggesting an association between the inferred cognitive ranking and absolute brain mass.

5 Discussion

5.1 Prior model

As seen in Figure 9, the consensus matrix recovers the empty order more accurately than the total order, as measured by the Hamming distance from the true partial order. This difference may be explained in part by the lack of marginal consistency of the uniform prior on $\mathcal{H}_M$ with respect to poset depth: as m increases, the prior increasingly favors partial orders of lower depth, as mentioned in Section 2.1. This motivates considering alternative prior models that

provide greater control over the structural properties of the underlying partial order. As we mentioned in Section 2.1 some probable alternatives include the ones suggested in Nicholls et al. [2025], but they have not been considered this time for the simplicity of the report.

5.2 Thank you!

I would like to thank Geoff for supervising the summer project and everyone who I had meaningful conversation about this project with. This has been a fruitful summer for me and I really enjoyed doing this project despite the ups and downs. If there are any questions about the project, please feel free to drop me an email: oweny0207@gmail.com

5.3 Further questions

Given more time, skill, and motivation, I would also like to investigate the followings:

1. Investigate how N scales with the magnitude of n . For instance it would be nice to know something like how much N should be increased by when n goes up by 1 in order to keep $Q_q^{(N)} = h_0$ with probability $1 - \eta$ for some $\eta \in (0, 1)$.
2. Investigate how changing r would affect the diffusion rate of the chain in Algorithm 2.
3. Develop a tractable, more efficient partial order proposal in the MCMC. The current one diffuses slowly and takes many steps to travels to different probability states.
4. Look for non-ignorable choices of bucket generation process and investigate how assuming them being ignorable would affect the posterior.
5. Investigate or develop another prior for the posterior that is less informative over the depth of the partial order.
6. Develop theories about how the Hamming distance between the q -consensus matrix changes with the depth of the true partial order, or the number of relations there are, when N is finite. Our experiments in Figure 6 suggests smaller q predicts h_0 when the depth of the true order is greater.

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6 Appendix

A Additional Results for Ranking Likelihoods

A.1 Results on the Uniform Likelihood

The following results are reproduced from an unpublished summer-project report by Sofia Monarchi. As the report is not publicly available, we include the statement and argument here for completeness; this result is not claimed as an original contribution of the present report.

Recall the partial-order Mallows model

$$p(y \mid h, \theta) = \frac{\exp\{-\theta d(y, h)\}}{Z(\theta, h)},$$

where

$$Z(\theta, h) = \sum_{z \in \mathcal{P}_M} \exp\{-\theta d(z, h)\},$$

and $d(y, h)$ is the Hamming distance between the relation matrices of the total order y and the partial order h .

Claim A.1. Let

$$d_{\min}(h) := \min_{y \in \mathcal{P}_M} d(y, h).$$

Then

$$d(y, h) = d_{\min}(h) \iff y \in \mathcal{L}[h].$$

Consequently,

$$\lim_{\theta \rightarrow \infty} p(y \mid h, \theta) = \frac{\mathbb{I}_{\{y \in \mathcal{L}[h]\}}}{|\mathcal{L}[h]|}.$$

Thus, the uniform linear-extension likelihood is obtained as the zero-noise limit of the partial-order Mallows model.

Proof. Let r denote the number of relations in h . Since y is a total order, its relation matrix contains exactly $\binom{|M|}{2}$ non-zero entries. Hence

$$d(y, h) \geq \binom{|M|}{2} - r.$$

Equality holds precisely when every relation in h is also present in y , which is equivalent to y being a linear extension of h . Therefore,

$$d_{\min}(h) = \binom{|M|}{2} - r,$$

and

$$\{y \in \mathcal{P}_M : d(y, h) = d_{\min}(h)\} = \mathcal{L}[h].$$

Now write

$$Z(\theta, h) = e^{-\theta d_{\min}(h)} \sum_{z \in \mathcal{P}_M} e^{-\theta(d(z, h) - d_{\min}(h))}.$$

It follows that

$$p(y \mid h, \theta) = \frac{e^{-\theta(d(y, h) - d_{\min}(h))}}{\sum_{z \in \mathcal{P}_M} e^{-\theta(d(z, h) - d_{\min}(h))}}.$$

As $\theta \rightarrow \infty$, all terms for which $d(z, h) > d_{\min}(h)$ vanish, while the terms corresponding to $z \in \mathcal{L}[h]$ are equal to one. Hence

$$\lim_{\theta \rightarrow \infty} p(y \mid h, \theta) = \begin{cases} \frac{1}{|\mathcal{L}[h]|}, & y \in \mathcal{L}[h], \\ 0, & y \notin \mathcal{L}[h]. \end{cases}$$

This is exactly the uniform linear-extension likelihood. □

Recall that under the uniform linear-extension likelihood,

$$p(y \mid h) = \frac{\mathbb{I}_{\{y \in \mathcal{L}[h]\}}}{|\mathcal{L}[h]|}.$$

The difficulty in using this likelihood within a Metropolis–Hastings algorithm is that evaluating $|\mathcal{L}[h]|$ is computationally intractable in general.

Suppose the current partial order is h and we propose $h' \sim q(\cdot \mid h)$. Introduce an auxiliary total order

$$y' \sim p(\cdot \mid h') = \text{Unif}(\mathcal{L}[h']).$$

The single-variable exchange acceptance probability is

$$\alpha(h' \mid h) = 1 \wedge \frac{\pi(h') p(y \mid h') p(y' \mid h) q(h \mid h')}{\pi(h) p(y \mid h) p(y' \mid h') q(h' \mid h)}.$$

Substituting the uniform linear-extension likelihood gives

$$\begin{aligned} \alpha(h' \mid h) &= 1 \wedge \frac{\pi(h') \frac{\mathbb{I}_{\{y \in \mathcal{L}[h']\}}}{|\mathcal{L}[h']|} \frac{\mathbb{I}_{\{y' \in \mathcal{L}[h]\}}}{|\mathcal{L}[h]|} q(h \mid h')}{\pi(h) \frac{\mathbb{I}_{\{y \in \mathcal{L}[h]\}}}{|\mathcal{L}[h]|} \frac{\mathbb{I}_{\{y' \in \mathcal{L}[h']\}}}{|\mathcal{L}[h']|} q(h' \mid h)} \\ &= 1 \wedge \frac{\pi(h') q(h \mid h') \mathbb{I}_{\{y \in \mathcal{L}[h']\}} \mathbb{I}_{\{y' \in \mathcal{L}[h]\}}}{\pi(h) q(h' \mid h) \mathbb{I}_{\{y \in \mathcal{L}[h]\}} \mathbb{I}_{\{y' \in \mathcal{L}[h']\}}}. \end{aligned} \quad (10)$$

Since y is drawn under the current state h and y' is drawn under the proposed state h' , we have

$$y \in \mathcal{L}[h], \quad y' \in \mathcal{L}[h']$$

by construction. Hence

$$\alpha(h' \mid h) = 1 \wedge \frac{\pi(h') q(h \mid h')}{\pi(h) q(h' \mid h)} \mathbb{I}_{\{y \in \mathcal{L}[h']\}} \mathbb{I}_{\{y' \in \mathcal{L}[h]\}}.$$

Thus the factors $|\mathcal{L}[h]|$ and $|\mathcal{L}[h']|$ cancel from the acceptance probability, so the MCMC update does not require the number of linear extensions to be evaluated.

A.2 MLE of the FM Likelihood

Suppose $H = (h_1, h_2, \dots, h_K)$ are K posets on the same ground set M . The intersection order of H is the partial order

$$\left(M, \bigcap_{i=1}^K \prec_{h_i} \right),$$

written $\bigcap_{i=1}^K h_i$. Jiang and Nicholls [2026] have shown that an MLE under the uniform likelihood is given by the intersection order of the observed total orders. We show that this is also the case for the FM likelihood.

Claim A.2. Given total-order data $l = (l^{(1)}, \dots, l^{(N)})$ on the same ground set $M = \{1, \dots, n\}$, let

$$L := \bigcap_{i=1}^N l^{(i)}$$

be their intersection order. Then L is an MLE under the FM likelihood. Under a uniform prior on $\mathcal{H}_M$, L is also a posterior mode.

Proof. The idea in this proof is pretty much the same as the one in Jiang and Nicholls [2026]. Let $h \in \mathcal{H}_M$. Recall that the likelihood is given by

$$p(l \mid h) = \prod_{i=1}^N \prod_{j=1}^n \frac{\mathbb{I}_{[l^{(i)} \in \mathcal{L}[h]]}}{|\max(h[l_{j:n}^{(i)}])|}.$$

If h does not agree with all of $l = (l^{(1)}, \dots, l^{(N)})$, then there exists some i such that $l^{(i)} \notin \mathcal{L}[h]$, and hence the likelihood vanishes. Such an h therefore cannot be an MLE.

Now suppose that h agrees with every $l^{(i)}$. By the definition of the intersection order L , every relation in h must also be present in L , and hence

$$\prec_h \subseteq \prec_L.$$

Indeed, if some relation in $\prec_h$ were not in $\prec_L$, then it would contradict at least one of the observed total orders.

For any $i \in \{1, \dots, N\}$ and $j \in \{1, \dots, n\}$, consider the remaining ground set $l_{j:n}^{(i)}$. Since $\prec_h \subseteq \prec_L$, adding the additional relations contained in L can only reduce the set of maximal elements. Therefore,

$$\max(L[l_{j:n}^{(i)}]) \subseteq \max(h[l_{j:n}^{(i)}]),$$

and hence

$$|\max(L[l_{j:n}^{(i)}])| \leq |\max(h[l_{j:n}^{(i)}])|.$$

Since both h and L agree with every observed total order, all of the indicator functions in their likelihoods are equal to one. It follows that

$$\frac{1}{|\max(h[l_{j:n}^{(i)}])|} \leq \frac{1}{|\max(L[l_{j:n}^{(i)}])|}$$

for every i and j . Multiplying over all i and j gives

$$p(l \mid h) \leq p(l \mid L).$$

Therefore L is an MLE under the FM likelihood. Under a uniform prior on $\mathcal{H}_M$, maximizing the posterior is equivalent to maximizing the likelihood, so L is also a posterior mode. $\square$

A.3 Failure of Projective Consistency for FM

We provide an example where the FM likelihood is not projectively consistent. Suppose $M = [3]$ and $h \in \mathcal{H}_M$ has relation set $\prec_h = \{(1 \prec 2)\}$. Now

$$\mathcal{L}[h] = \{(1 \succ 3 \succ 2), (3 \succ 1 \succ 2), (1 \succ 2 \succ 3)\}.$$

Notice under FM we observe these 3 linear extensions with probability $1/4$, $1/2$ and $1/4$ respectively. Marginalizing over actor 1 we then see that we observe $(3 \succ 2)$ with probability $3/4$ and $(2 \succ 3)$ with probability $1/4$, which is not $1/2$ as what FM likelihood would give. $\square$

The intuition here is that 3 has a higher chance to be above 2 since it is in the set of maximal elements more often than 2.

A.4 Ignorability is More General than a Common Weight Distribution

Fix $o = M = \{1, 2, 3\}$. We give an example of an ignorable g for which there does not exist a random variable D on $\mathcal{B}_o$ such that

$$\sigma_y(B) \sim D \quad \text{for all } y \in \mathcal{P}_o,$$

where $B \mid y, o \sim g(\cdot \mid y, o)$ and D is independent of $B \mid y, o$.

We write a bucket order in the form

$$[a_1^{(1)}, \dots, a_1^{(B_1)}] \succ \dots \succ [a_1^{(k)}, \dots, a_1^{(B_k)}],$$

where $a_i^{(j)}$ denotes the i th actor in the j th bucket and B_j is the size of the j th bucket.

Let

$$b_1 = 1 \succ [2, 3].$$

For an ignorable g and $b \in \mathcal{B}_o$, let c_b denote the common value of $g(b \mid y, o)$ over all $y \in \mathcal{L}[b]$. For a total order $b = a_1 \succ a_2 \succ a_3 \in \mathcal{P}_o$, write this value as $c_{a_1 a_2 a_3}$.

Define g by setting

$$c_{b_1} = \frac{1}{2}, \quad c_b = 0 \quad \text{for all } b \in \mathcal{B}_o \setminus \mathcal{P}_o, \quad b \neq b_1,$$

and

$$c_{123} = c_{132} = \frac{1}{2}, \quad c_{312} = c_{321} = c_{213} = c_{231} = 1.$$

Then $g(\cdot \mid y, o)$ is a probability distribution for every $y \in \mathcal{P}_o$, and g is ignorable by construction.

Suppose, for a contradiction, that there exists a random variable D on $\mathcal{B}_o$ such that

$$\sigma_y(B) \sim D \quad \text{for every } y \in \mathcal{P}_o.$$

Consider $y = 1 \succ 2 \succ 3$. Since $b_1 = 1 \succ [2, 3] \in \mathcal{L}[b_1]$,

$$\mathbb{P}[D = 1 \succ [2, 3]] = g(b_1 \mid y, o) = \frac{1}{2}.$$

Now consider $y' = 3 \succ 2 \succ 1$. Since $3 \succ [2, 1] \in \mathcal{B}_o$ and $c_{3 \succ [2, 1]} = 1$,

$$\mathbb{P}[D = 1 \succ [2, 3]] = g(3 \succ [2, 1] \mid y', o) = 1.$$

This is a contradiction, so no such random variable D exists. $\square$

A.5 An NASC for being ignorable

We give a characterization of ignorability for a fixed observed subset $o \subseteq M$, together with a useful corollary.

Claim A.3. Fix $o \subseteq M$. For each $b \in \mathcal{B}_o$, let $c_b \geq 0$. The collection $c = (c_b)_{b \in \mathcal{B}_o}$ uniquely defines an ignorable bucket-observation mechanism $g(\cdot \mid y, o)$ if and only if

$$\sum_{b: y \in \mathcal{L}[b]} c_b = 1, \quad \forall y \in \mathcal{P}_o. \quad (11)$$

In particular, the set of all such collections c is a convex polytope. Consequently, if g_1 and g_2 are ignorable, then $\lambda g_1 + (1 - \lambda)g_2$ is also ignorable for every $\lambda \in [0, 1]$.

Proof. If g is ignorable, then by **(A2)** there exists, for each $b \in \mathcal{B}_o$, a constant c_b such that

$$g(b \mid y, o) = c_b \mathbb{I}_{\{y \in \mathcal{L}[b]\}}.$$

Since $g(\cdot \mid y, o)$ is a probability distribution for every $y \in \mathcal{P}_o$, normalization gives

$$\sum_{b: y \in \mathcal{L}[b]} c_b = 1.$$

Conversely, suppose $c_b \geq 0$ satisfies (11), and define

$$g(b \mid y, o) := c_b \mathbb{I}_{\{y \in \mathcal{L}[b]\}}.$$

Equation (11) ensures that $g(\cdot \mid y, o)$ is a probability distribution. It satisfies **(A2)** by construction and **(A1)** since it does not depend on h . Hence g is ignorable.

For the corollary, enumerate $\mathcal{P}_o = \{y_i\}_{i \in I}$ and $\mathcal{B}_o = \{b_j\}_{j \in J}$, and define

$$A_{ij} = \mathbb{I}_{\{y_i \in \mathcal{L}[b_j]\}}.$$

The set of ignorable mechanisms is therefore identified with

$$\{c \in \mathbb{R}^{|J|} : c \geq 0, \quad Ac = \mathbf{1}\},$$

which is a convex polytope. The convex-combination result follows immediately. $\square$

A.6 Proof of Lemma 4

Proof. For $i = 1, \dots, N$ and $h \neq h_0$, define

$$Z_i(h) := \log \left(\frac{P_h(X_i)}{P_{h_0}(X_i)} \right).$$

For fixed $h \neq h_0$, the random variables $Z_1(h), \dots, Z_N(h)$ are i.i.d. with

$$\mathbb{E}_{h_0}[Z_i(h)] = -D_{\text{KL}}(P_{h_0} \parallel P_h) = -c_h.$$

Since $\mathcal{X}$ is finite and $P_h(x) > 0$ for all $x \in \mathcal{X}$, the range of $Z_i(h)$ is bounded by R . Hence, by Hoeffding's inequality,

$$\begin{aligned} \mathbb{P}_{h_0} \left[\frac{1}{N} \sum_{i=1}^N Z_i(h) > -\frac{c_h}{2} \right] &= \mathbb{P}_{h_0} \left[\frac{1}{N} \sum_{i=1}^N (Z_i(h) + c_h) > \frac{c_h}{2} \right] \\ &\leq \exp \left(-\frac{c_h^2}{2R^2} N \right) \\ &\leq \exp \left(-\frac{\Delta^2}{2R^2} N \right). \end{aligned} \tag{12}$$

Applying the union bound over all $h \neq h_0$ gives

$$\begin{aligned} \mathbb{P}_{h_0} \left[\exists h \neq h_0 : \frac{1}{N} \sum_{i=1}^N Z_i(h) > -\frac{c_h}{2} \right] \\ \leq \sum_{h \neq h_0} \mathbb{P}_{h_0} \left[\frac{1}{N} \sum_{i=1}^N Z_i(h) > -\frac{c_h}{2} \right] \\ \leq (|\mathcal{H}_M| - 1) \exp \left(-\frac{\Delta^2}{2R^2} N \right). \end{aligned} \tag{13}$$

It remains to show that, for sufficiently large N , the complement of the event in (13) implies $Q_q^{(N)} = h_0$.

Suppose that

$$\frac{1}{N} \sum_{i=1}^N Z_i(h) \leq -\frac{c_h}{2} \quad \forall h \neq h_0.$$

For any $h \neq h_0$, the posterior odds of h against h_0 satisfy

$$\begin{aligned} \frac{\Pi_N(h)}{\Pi_N(h_0)} &= \frac{\pi(h)}{\pi(h_0)} \prod_{i=1}^N \frac{P_h(X_i)}{P_{h_0}(X_i)} \\ &= \frac{\pi(h)}{\pi(h_0)} \exp \left(\sum_{i=1}^N Z_i(h) \right). \end{aligned} \tag{14}$$

Therefore,

$$\frac{\Pi_N(h)}{\Pi_N(h_0)} \leq \frac{\pi(h)}{\pi(h_0)} \exp \left(-\frac{c_h}{2} N \right) \leq \frac{\pi(h)}{\pi(h_0)} \exp \left(-\frac{\Delta}{2} N \right).$$

Recall that

$$\epsilon_N = 1 - \Pi_N(h_0).$$

Since $\Pi_N(h_0) > 0$,

$$\begin{aligned}\epsilon_N &\leq \frac{\epsilon_N}{1 - \epsilon_N} = \sum_{h \neq h_0} \frac{\Pi_N(h)}{\Pi_N(h_0)} \\ &\leq \sum_{h \neq h_0} \frac{\pi(h)}{\pi(h_0)} \exp\left(-\frac{\Delta}{2}N\right) \\ &= A \exp\left(-\frac{\Delta}{2}N\right).\end{aligned}\tag{15}$$

If

$$N > \frac{2}{\Delta} \log\left(\frac{A}{E_q}\right),$$

then

$$A \exp\left(-\frac{\Delta}{2}N\right) < E_q.$$

Hence $\epsilon_N < E_q$, and by (9),

$$Q_q^{(N)} = h_0.$$

Thus,

$$\{Q_q^{(N)} \neq h_0\} \subseteq \left\{ \exists h \neq h_0 : \frac{1}{N} \sum_{i=1}^N Z_i(h) > -\frac{c_h}{2} \right\}.$$

Taking probabilities and applying (13) gives

$$\mathbb{P}_{h_0}(Q_q^{(N)} \neq h_0) \leq (|\mathcal{H}_M| - 1) \exp\left(-\frac{\Delta^2}{2R^2}N\right),$$

which completes the proof. $\square$

B Additional Simulation Results

B.1 Runtime of the chain at different n and N

n	$N = 10$	$N = 20$	$N = 30$
5	1.183 ± 0.021	1.624 ± 0.038	1.856 ± 0.074
10	1.338 ± 0.036	1.616 ± 0.059	2.069 ± 0.090
15	1.710 ± 0.039	2.165 ± 0.066	1.997 ± 0.076
20	1.899 ± 0.064	2.607 ± 0.104	2.949 ± 0.138
25	2.832 ± 0.104	2.189 ± 0.092	3.388 ± 0.154
30	3.285 ± 0.138	3.607 ± 0.179	2.643 ± 0.156
35	2.617 ± 0.100	4.221 ± 0.176	4.586 ± 0.201
40	4.496 ± 0.177	3.194 ± 0.184	3.489 ± 0.223

Table 1: Mean runtime (seconds) with 95% confidence intervals.

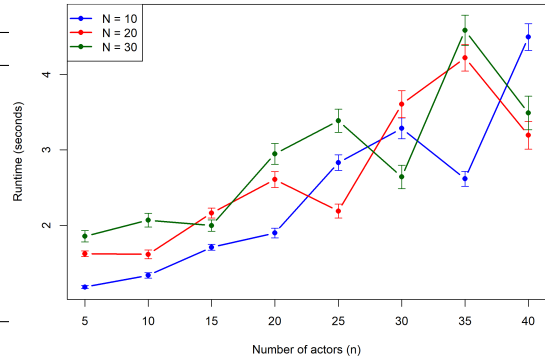


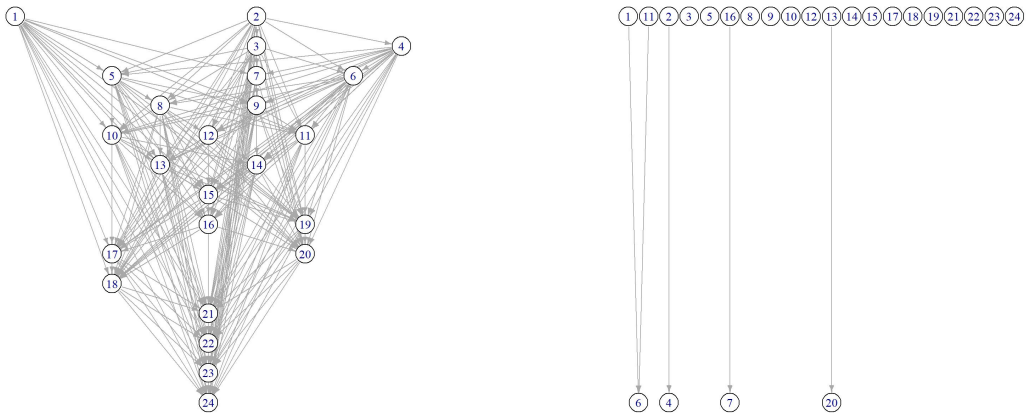
Figure 8: Mean runtime for different values of n and N . Error bars show 95% confidence intervals.

To investigate the runtime of the MCMC, we ran the algorithm on our computer for $n = 5, 10, 15, \dots, 40$ and $N = 10, 20, 30$, with 10^4 steps per run. For each combination of n

and N , we repeated the experiment 100 times, targeting a randomly generated poset on n actors. We then calculated the 95% CI of the runtime for each case, as shown in Table 1 and Figure 8. The algorithm generally takes more time to run as N increases, since there are more terms to evaluate in the likelihood, although this effect appears to be small compared with the effect of increasing n . The runtime also generally increases with n . Over the range considered here, the growth appears relatively moderate and is consistent with polynomial rather than exponential scaling, although experiments at larger values of n would be required to characterize the asymptotic scaling more precisely.

B.2 More diagrams

These are more the diagrams from Section 4.3



(a) Total-order case.

(b) Empty-order case.

Figure 9: The empirical .5-consensus orders for the total-order and empty-order cases, at $m = 24$.